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Behavioral and Hormonal Responses to Separation in Infant Rhesus Monkeys and Mothers

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Effects of social stimuli on behavioral and physiological responses were examined in infant rhesus monkeys at 4 and 9 months of age. Infants and mothers were removed from the social group and housed as dyads. Following this period, infants were removed and separated under four counterbalanced conditions: (a) totally isolated—placed in a holding cage for 24 hr; (b) mother present, no contact—housed in a single cage in view of their mother, no contact; (c) mother present, contact—similar to above, with mother in proximity to the infant; and (d) peer present—separated but in proximity to a peer. In the first experiment, the infants rarely vocalized when totally isolated but showed high rates of vocalization in the presence of the mother, both with and without contact. In the mother-present conditions, they failed to show a plasma cortisol response. In contrast, totally isolated infants showed a significant elevation in plasma cortisol. At 9 months of age, these infants were separated for 3 days under two different conditions: mother present and totally isolated. Once again, the infants that were totally isolated showed little vocalization but significant elevations in plasma cortisol. In contrast, infants separated in the presence of their mothers showed high vocalization rates but no cortisol response. The concepts of protest and despair are discussed as they relate to behavioral and physiological differences observed following different separation paradigms.

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Investigations of mother–infant separation in monkeys have found that infants typically respond to separation from mother initially with hyperactivity and increased vocalization ("protest" behavior) and, if the separation is prolonged, may then exhibit sharp decreases in activity and awareness ("despair" behavior; Mineka & Suomi, 1978). However, a number of variations on this typical response pattern have been reported, and one set of variables that has influenced infants' behavior while apart from the mother has to do with the separation environment. If substitute care is available to a separated infant, as with bonnet macaques (Rosenblum & Kaufman, 1968) and squirrel monkeys (Coe, Mendoza, Smotherman, & Levine, 1978) that were adopted by "aunts" after their mothers were removed, little or no behavioral distress may be evident.

Another aspect of the separation environment found to affect infant behavior is the degree of access to the mother during separation. The relation between infant distress and degree of maternal access is not linear. Infants separated from their mothers by a wire mesh partition, which allowed limited physical contact, showed only minimal distress behavior; however, when the members of the dyad were separated by a Plexiglas panel, which prevented tactile contact, the usual infant protest behavior was observed (Suomi, Collins, Harlow, & Ruppenthal, 1976). Seay and Harlow (1965) reported that infants with no access to the mother (the mothers were removed to a separate room) exhibited less intense protest than infants separated from the mother by Plexiglas. In our laboratory, isolating rhesus infants from their mothers

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and other conspecifics evoked less protest behavior than did separating them in view of the mother and/or social group (Levine, Franklin, & Gonzalez, 1984). Thus, visual contact with mother in the absence of tactile contact appears to elicit more vocalization in infants than either limited tactile contact or total isolation from the mother.

Behavioral measures, particularly vocalization, have often been used to indicate the degree of distress or arousal during separation. However, the activity of the pituitary-adrenal system has also been shown to be a sensitive index of physiological arousal (Hennessy & Levine, 1979) and to respond to separation in infant monkeys (Coe et al., 1978; Gunnar, Gonzalez, Goodlin, & Levine, 1981; Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978; Smotherman, Hunt, McGinnis, & Levine, 1979). In rhesus infants, elevations in plasma cortisol concentrations have been reported 30 min and 3 hr following separation from the mother (Gunnar et al., 1981; Smotherman et al., 1979) or from the mother and social group (Levine et al., 1984).

Physiological and behavioral indexes of infant arousal during separation have not always corresponded; for example, squirrel monkey infants that remained in their social groups after their mothers had been removed showed no protest behavior, yet they had elevated levels of plasma cortisol equal to those seen when they were housed alone during brief separation (Coe et al., 1978). In rhesus infants, as well, behavioral and adrenocortical responses to separation have not necessarily covaried. Gunnar, Gonzalez, and Levine (1980) found that, compared with being housed alone, having a peer in the same cage during separation from the mother and social group reduced an infant's behavioral distress but had no effect on elevations in plasma cortisol concentration.

In the present studies with rhesus infants, we utilized both behavioral and adrenocortical measures to evaluate the effects of degree of access to the mother on the psychobiological response to separation.

Experiment 1

Method

Subjects. The subjects of the first experiment were 8 infant rhesus monkeys (2 females and 6 males, aged 4–6 months) and their mothers. The animals lived in two established social groups, each housed in an outdoor enclosure (5 × 10 × 3 m) with an adjoining indoor area (3 × 5 × 3 m). Animals received standard laboratory monkey chow and fresh fruit; water was freely available. The study was conducted during September and October.

Housing. In order to stabilize baseline cortisol levels, two mother–infant dyads were captured from their social group at 1300 hours, and each dyad was housed in a standard laboratory cage (64 × 74 × 38 cm) in the same room for a 5-day adaptation period. On Day 6, the infants were removed from the cages at 1300 hours and were each placed in an identical cage under one of four conditions: (a) isolation (ISO)—alone in a separate room out of any contact with other rhesus monkeys; (b) peer adjacent (PEER)—with the other separated infant in an adjacent cage, and the mothers removed to a separate area, out of any contact; (c) mother adjacent (ADJ)—with its mother in an adjacent cage (approximately 25 cm away); and (d) mother tactile (TAC)—with its mother in a nearby cage (approximately 4 cm away), which allowed limited tactile contact. During the ADJ and TAC conditions, the other mother–infant dyad was in the same room in the same condition. During ISO and PEER conditions, the mothers were each individually caged in isolation from other rhesus monkeys. Separation lasted 24 hr, and infants were then reunited with their mothers for 24 hr. This separation–reunion procedure was repeated until each infant had experienced all four separation conditions. After the last reunion, the dyads were returned to their social group. The entire procedure was repeated with different mother–infant dyads until all eight dyads had been tested. The order of separation conditions was counterbalanced across subjects.

Behavioral observations. A trained observer sat behind a screen in front of the infant's cage and observed each infant for 30 min in the morning (between 0830 and 0930) and 30 min in the afternoon (between 1400 and 1500) on the fifth adaptation day (BASE) and on each separation day. The following behaviors were recorded on checksheets divided into 30-s time intervals: (a) vocalization—total frequency of whoo calls and screams; (b) movement—number of seconds moving (changing location; recorded with a stopwatch); (c) cage biting, stereotypy (repetition of a stylized behavior pattern at least three times), eating, and drinking—each recorded as occurrence/nonoccurrence for each interval. Behavior recorded during BASE conditions and TAC separations also included (d) contact with mother—recorded "on-the-signal" every 30 s; and (e) grooming and play—recorded as occurrence/nonoccurrence for each interval.

Plasma cortisol determination. Blood samples (0.5 ml) were drawn by saphenous venipuncture. Subjects were manually restrained during the 1–5 min required

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to take the sample. Blood was collected in heparinized syringes and centrifuged, and the plasma was frozen at −20°C until assayed. Cortisol levels were measured by radioimmunoassay, according to the procedures of Klemm and Gupta (1975). Blood samples were drawn from each animal at 1600 hours on the day of capture, the next day, on the fifth day of adaptation (3, 27, and 99 hr after capture, respectively), and on each day of separation (3 hr after separation).

Results

An analysis of variance (ANOVA) and post hoc Newman-Keuls tests of the plasma cortisol levels of mothers and infants during adaptation (Figure 1) revealed significant effects of age (mother vs. infant), $F(1, 14) = 53.50$, $p < .001$, and time after capture, $F(2, 28) = 22.98$, $p < .001$, as well as a significant interaction, $F(2, 28) = 10.69$, $p < .001$. Mothers' cortisol levels were higher ($p < .05$) than the infants' levels, particularly in their initial response to capture. The day after capture, mothers, but not infants, showed a significant reduction ($p < .05$) in cortisol; by the fifth day after capture, both mothers and infants had significantly lower ($p < .05$) cortisol levels. The cortisol levels on the day before separation were used as the base values for comparison with the separation cortisol levels.

Marked differences in infant cortisol levels were found to be dependent upon the mother's presence during separation (Figure 2). Analysis of variance of infant cortisol levels, and subsequent post hoc Newman-Keuls tests, revealed significantly lower cortisol during the ADJ and TAC conditions than during the ISO and PEER separations, $F(4, 28) = 4.83$, $p < .005$. The cortisol levels during the ISO and PEER conditions were equivalent and significantly elevated over basal levels ($p < .05$), whereas the ADJ and the TAC separations produced only small changes in cortisol which did not differ from basal levels. Thus, separation in the mother's presence, with or without tactile access, was not accompanied by elevation in plasma cortisol.

Behavioral observations. The infants in this experiment were alert and continued to eat and drink throughout all separations. Mothers and infants spent only 3% of the observation intervals in contact during the TAC separation, compared with 78% of intervals during the basal observations. The infants emitted whoo calls and screams at a significantly higher rate during the ADJ and TAC separations than the ISO and PEER separations, $F(3, 21) = 5.34$, $p < .01$ (Figure

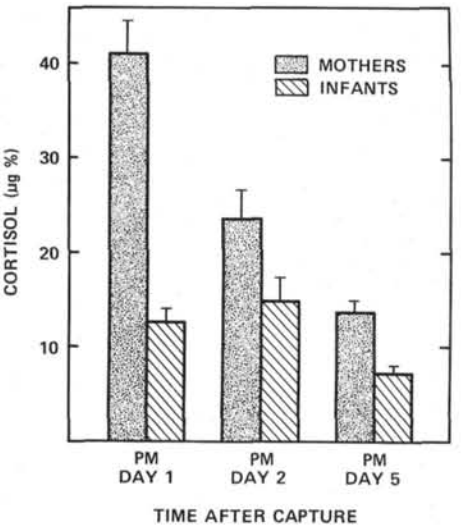


Figure 1. Mean (± *SE*) plasma cortisol of mothers and infants at different times following removal from the social group.

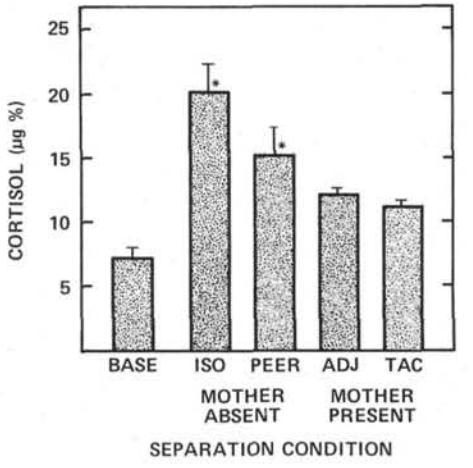


Figure 2. Mean (± *SE*) plasma cortisol under different separation conditions. (Asterisks indicate significant differences from basal, adjacent [ADJ], and tactile [TAC] conditions; isolated [ISO] and peer [PEER] did not differ.)

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3A). No significant differences were found in vocalization rates between the ADJ and TAC, or between the ISO and PEER, conditions. Vocalization decreased significantly during the course of the separation interval; vocalization was higher at 1 hr after separation than the next morning, $F(1, 7) = 29.97$, $p < .01$. The infants moved significantly more during the ISO, PEER, and ADJ conditions than during the basal observations, $F(4, 28) = 10.30$, $p < .01$. Differences in moving time between ISO and PEER, and between ADJ and TAC, were not significant, and there were no differences in movement 1 hr after separation when compared with that of the next morning, $F(1, 7) = 0.07$, *ns* (Figure 3B). Cage biting, which is presumably another measure of infant agitation, changed with the conditions, $F(4, 28) = 3.07$, $p < .05$ (Figure 3C). The only significant difference (Newman-Keuls, $p < .05$) was between the BASE and the PEER separation. Also, cage biting occurred more often the next morning than 1 hr after separation, $F(1, 7) = 11.84$, $p < .02$. Although stereotypic behaviors may also be indicative of infant agitation, they were seen too infrequently in these studies to allow statistical analysis. Three infants exhibited stereotypies: 2 in all separations and 1 in only the ISO and ADJ conditions.

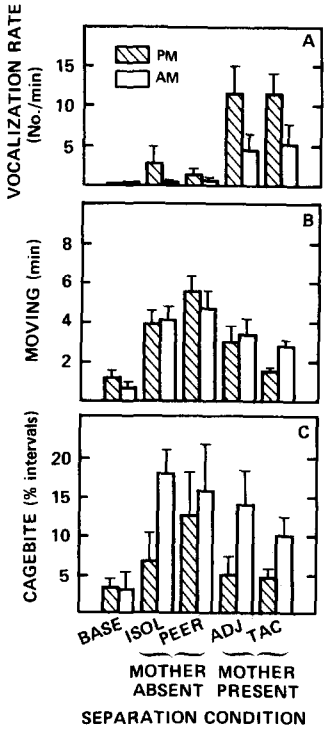


Figure 3. Behavioral responses following separation under different separation conditions. (Bars represent means ± *SE*. ISOL = isolated; ADJ = adjacent; TAC = tactile.)

Discussion

The findings of this study are consistent with those of several other studies which found that the behavioral sequelae following disruption of mother–infant relations are extremely dependent upon the environmental conditions that prevail at the time of separation. Infants with visual or tactile access to the mother showed a significantly higher number of vocalizations; and, in fact, these vocalizations were emitted almost exclusively in the presence of the mother during separation. Indications of relatively low vocalization rate have been observed before in separated, isolated infants relative to those in contact with social stimuli. Hinde and McGinnis (1977) reported that whoo calls occurred at the rate of 3.6 per minute in isolated infants and 6 per minute in infants that remained in their social groups after their mothers had been removed. Levine et al. (1984) reported only 1 whoo call per minute in isolated infants but nearly 9 per minute for infants within visual and auditory contact of the social group. The absence of the mother in the social group in that experiment tended to elicit even greater vocalizations for the first 2 days following separation. Therefore, it seems apparent that, rather than being an invariant component of infant behavior following separation, these so-called distress vocalizations are elicited by specific social stimuli during separation from the mother. If whoo calls function to reestablish contact with the mother, it is reasonable to expect infants to emit the highest rate of vocalizations at a time when they are most likely to be effective, that is, in the presence of stimuli associated with maternal contact. Visual or auditory access to the mother or social group is apparently a sufficient stim-

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ulus, whereas similar access to a single separated infant is not.

In contrast to the behavioral data, infant rhesus monkeys separated from their mothers showed no increase in pituitary-adrenal activity when the separation took place in the presence of the mother. However, the infants that were completely isolated and those separated in the presence of a single peer showed significant elevations of plasma cortisol following separation, although vocalization occurred at a significantly lower rate. These data appear to be paradoxical. Infants that are vocalizing in the presence of their mothers do not show an endocrine response, whereas infants that appear to be much less behaviorally agitated show a striking physiological response indicative of stress. It appears from these data that there is an inverse relation between the endocrine state and the behavioral state. However, it is not valid to conclude that this inverse relation exists under all conditions in all primate species. Squirrel monkeys separated from their mothers, but permitted to remain in the social group with peers and aunts, show few of the behavioral responses usually associated with separation; however, they do show a significant elevation of cortisol (Coe et al., 1978). These same animals show much higher levels of vocalization when totally separated from the mother, but they also show an equivalent cortisol response. Thus, the inverse relation we observed in the rhesus macaque in this experiment does not appear to be present under all conditions in other primate species.

When dealing with both behavioral and physiological responses to separation, the temporal aspects of these responses become very important dimensions. The so-called "protest" and "despair" responses to separation follow a time course that shows an initial high rate of vocalization, followed by a decline in vocalization and a decline in movement. If one extends separation to a 24-hr period, the cortisol response of the squirrel monkey also shows an inverse relation. Thus, at the end of 24 hr, totally isolated infants show almost no vocalizations but have significantly elevated cortisol values over those observed during briefer periods of separation (Coe, Wiener, & Levine, 1983).

The role of the cortisol response in relation to emitted behavior has not yet come under serious investigation. In an experiment with a similar paradigm, Suomi et al. (1976) reported data similar to those in this experiment, namely, that totally isolated animals showed reduced vocalization compared with animals in the presence of their mothers when tactile access was not available. Those data were interpreted as indicating that the response to separation was diminished if the cues were not available, and they led to what is termed the "out-of-sight, out-of-mind" hypothesis. The data presented in this study lead to a very different interpretation. The absence of maternal cues appears to result in a more physiologically aroused state, although vocalizations were diminished significantly.

We are faced with the question of what these different variables may indeed be measuring. In both the human and the primate literature, vocalization has been the predominant measure of distress. Terms such as *protest* and *distress vocalization* have come into common usage. Most researchers studying the response to separation have considered protest to be the universal initial reaction to maternal separation for primate infants, probably the product of natural selection (Kaufman & Rosenblum, 1967; Rosenblum & Plimpton, 1981). However, the data from this study and from other studies in the literature indicate that vocalization is by no means an invariant response to separation and that it appears to be extremely sensitive to environmental contraints. In a previous article we did, in fact, question the use of the term *protest* as an adequate description of the infant's vocalization response to separation (Levine et al., 1984). *Protest* has been defined, in primate species at least, as the initial period following separation that is characterized by more frequent vocalization and agitated movements. Current theories of attachment would lead one to predict that the presence or absence of social stimuli should not in any way affect this protest response, and, in fact, attachment theory would have predicted much

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greater vocalization in the absence of the mother than in the mother's presence—opposite to the result reported here. We hypothesize that, rather than being an index of distress and behavioral agitation, vocalizations are, in fact, the major signal that the infant uses under normal social conditions to achieve proximity with the mother. This concept leads to the prediction that when appropriate social stimuli are present, the infant's attempts to achieve proximity using vocalizations would be more vigorous than when the stimuli were not available.

It was surprising that infants separated in the presence of their mother failed to show a cortisol response. This is the first time in our laboratory that separation did not lead to some level of activation of the pituitary-adrenal system. One hypothesis that may account for this finding is that vocalization serves a coping function and thereby eliminates the cortisol response that would normally follow loss of the mother. Rosenblum and Plimpton (1981) hypothesized that one possible function of high rates of vocalization observed during the first day of separation may be an attempt by the infant to reestablish contact with the mother. Thus, as long as the stimuli that elicit vocalization are present, the infant continues to vocalize. These vocalizations may represent successful coping responses which, in turn, reduce the level of physiological arousal.

Experiment 2

The purpose of the second experiment was to further examine the factors that altered the behavioral and physiological response to differing separation environments. In the initial experiment, the separations were brief (24 hr), and the absence of a corticoid response when the mother was present during separation might have been a result of the relatively short interval of time (3 hr) between the onset of the separation and obtaining the blood sample. If the separation were extended, there may indeed be changes in the infants' physiological responses despite the presence of the mother. During the first experiment, we also noted that the values of plasma cortisol, both during the basal conditions and following separations, were low compared with those in other studies we have performed on similar monkeys. However, the late afternoon is the nadir of the diurnal cortisol cycle in rhesus, and the fact that blood was collected at that time in the first experiment could account for the low values obtained. It has been reported that the magnitude of the response to separation, both behaviorally and in terms of autonomic responses, appears to be greatest the morning after separation (Reite & Short, 1978). Thus, in the second experiment, we collected blood samples both in the morning and in the afternoon to determine whether there was a difference in the response to separation as a function of circadian rhythm.

Method

Subjects. Subjects in this experiment were eight mother–infant dyads; the infants were approximately 8–10 months of age. Seven of these mother–infant dyads had been used in the first experiment. The additional dyad used was from the same social groups but had no prior separation history. This experiment was conducted between January and April.

Housing. Two mother-infant dyads were captured at 1300 hours and housed as in Experiment 1 for a 4-day adaptation period. Infants were removed from their mothers on Day 5 and at 1300 hours were placed in adjacent identical cages under one of two conditions: (a) mother present (MoPrs)—mothers remained in an adjacent cage (approximately 25 cm away) and (b) mother absent (MoAbs)—mothers were removed to a separate area out of contact with the infants. During the latter condition, mothers were caged adjacent to each other approximately 1 m away from the social group. Separation lasted 3 days, after which the infants were reunited with the mother for 4 days and then separated into the other condition for 3 subsequent days. After the second separation, infants and mothers were returned to their social groups. The entire procedure was repeated with different dyads until all eight dyads had been tested. The order of separation conditions was balanced across the subjects.

Behavioral observations. Each infant was observed for 20 min between 1400 and 1500 hours on the day before (Day 0) and on each day of separation (Days 1, 2, and 3). Day 0 was considered a basal condition. Observation conditions and behaviors recorded were as in Experiment 1 except that infant movement was scored as one of three possible infant positions (moving, stationary but alert, or huddled) recorded on-the-signal every 30 s.

Plasma cortisol. Blood samples were obtained, processed, and analyzed as in Experiment 1. Blood was drawn at 1600 hours on the day of capture, at

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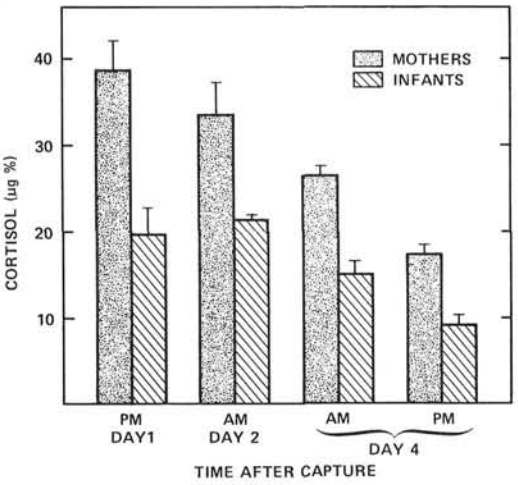


Figure 4. Mean (± *SE*) plasma cortisol of mothers and infants at different times following removal from the social group.

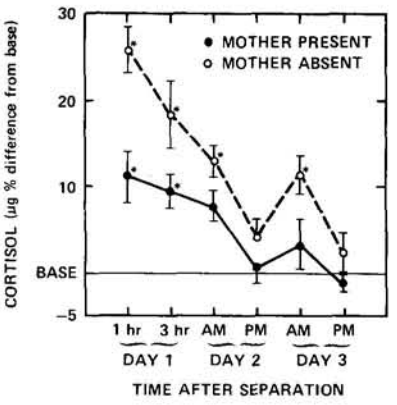


Figure 5. Mean (± *SE*) plasma cortisol responses for infants at different times following separation. (Asterisks indicate statistically significant differences from basal levels.)

0900 on the next day, and at 0900 and 1600 on the fourth adaptation day. During the separations, blood was drawn at 1400 and 1600 on the day of separation (Day 1; 1 and 3 hr after separation) and at 0900 and 1600 on each of the next 2 days (Days 2 and 3). Blood samples were also taken at 0900 and 1600 on the day before the second separation to once again establish basal cortisol levels.

Results

Adaptation. Similar to the data obtained in the first experiment, the mothers' cortisol values were higher than the infants' cortisol values during adaptation, $F(1, 14) = 35.99$, $p < .01$. Values on the morning after capture were not different from those 3 hr after capture for the mothers or infants, but both mother and infant had adapted to the new housing with significantly lower cortisol levels on the fourth day (Figure 4). As in the first experiment, cortisol levels from the day before separation were used as a basal sample for comparison with separation values. Mothers had higher cortisol levels than infants, and for both mothers and infants the morning levels were significantly higher than the afternoon levels. Levels at each time did not differ between the two basal days, so they were combined to yield mean basal morning and afternoon values. Mothers' basal cortisol concentrations were 21.6 μg% (morning) and 15.9 μg% (afternoon); comparable infant values were 13.8 μg% and 9.3 μg%.

Separation. The effect of the presence or absence of the mother at more extended times following separation also proved to be significant. An ANOVA revealed significantly higher cortisol levels during mother-absent than mother-present separation, $F(1, 7) = 32.96, p < .001$ (Table 1). In Figure 5, these cortisol concentrations at each

Table 1
Mean (±SE) Plasma Cortisol Concentrations in 8 Infant Monkeys at Various Intervals After Separation When the Mother Was Present or Absent

	Plasma cortisol (in μg%)					
	Day 1		Day 2		Day 3	
Condition	1 hr	3 hr	AM	PM	AM	PM
Mother present	20.5 ± 2.3	18.8 ± 2.2	21.6 ± 2.0	9.6 ± 1.1	17.1 ± 2.2	8.2 ± 1.1
Mother absent	35.1 ± 3.0	27.7 ± 3.6	26.8 ± 1.7	13.3 ± 1.9	25.2 ± 2.4	11.7 ± 1.8

Note. AM = 0900 hours; PM = 1600 hours.

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time point during separations are shown as differences from the appropriate morning or afternoon base concentration. Values on Day 1 (1 and 3 hr), Day 2 (afternoon) and Day 3 (afternoon) are compared with the afternoon base; Day 2 (morning) and Day 3 (morning) are compared with the morning base. Differences from the basal values were assessed with a Dunnett's t which revealed significant elevations ($p < .05$) in cortisol on Day 1, at 1 and 3 hr after separation in both MoPrs and MoAbs conditions. In the MoPrs condition, both morning and afternoon cortisol had returned to basal level on Days 2 and 3. During MoAbs separation, although afternoon cortisol levels were basal on Days 2 and 3, morning levels remained significantly elevated. Thus, in this experiment, infants showed increased adrenocortical activation over basal levels when separated in the presence of their mothers, but the magnitude and duration of the response were smaller than when the mother was absent during the period of separation.

The mothers failed to show an elevation of cortisol when separated in the presence of their infants. However, when separated from their infants and in the presence of the social group, there was an initial elevation of plasma cortisol on Day 1, 1 and 3 hr after separation. An ANOVA revealed a significant main effect of time, $F(5, 35) = 11.17, p < .01$, and a significant Conditions × Time interaction, $F(5, 35) = 3.76, p < .01$. Post hoc Newman-Keuls test demonstrated a significant difference ($p < .05$) only while the mothers were adjacent in the social group; values on Day 1 were higher ($p < .05$) than those on Days 2 and 3 (Figure 6).

Behavior. Vocalization was once again more frequent in the presence of the mother than when the mother was absent, $F(1, 7) = 9.80, p < .02$ (Figure 7A). Vocalization decreased significantly over the course of separation; vocalizations were emitted less frequently on Days 2 and 3 than on the initial day of separation, $F(2, 14) = 4.75, p < .05$. In addition, the 7 infants that had separation experience in the first experiment vocalized less often in the second experiment (12.5 calls per minute in Experiment 1; 5.7 calls per minute in Experiment 2). Infants moved more on all separation days than during basal observations, $F(3, 21) = 16.49, p < .001$, but there were no significant differences over days of sep-

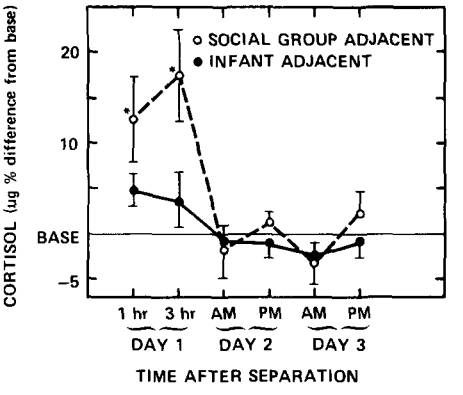


Figure 6. Mean (± *SE*) plasma cortisol responses for mothers at different times following separation. (Asterisks indicate statistically significant differences between social group adjacent and infant adjacent.)

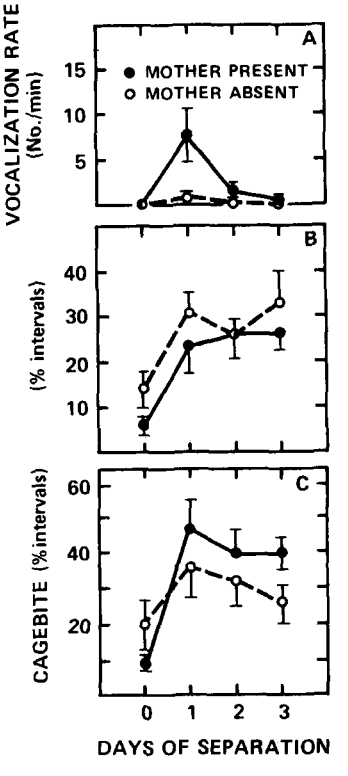


Figure 7. Behavioral responses following separation under different separation conditions. (Data are means ± *SE*.)

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aration, or between MoPrs and MoAbs conditions, $F(1, 7) = 3.05$, $p > .05$ (Figure 7B). Cage biting was more frequent during MoPrs than MoAbs separations, $F(1, 7) = 11.84$, $p < .02$. More cage biting occurred on Day 1 of separation than during basal observations ($p < .05$). On Days 2 and 3, the scores were not different from basal or Day 1 scores (Figure 7C). Stereotypic behavior was shown by only 1 infant, and this infant showed these behaviors in both separation conditions. In this experiment, as in the previous one, none of the infants showed any evidence of total cessation of movement or huddling postures; they were alert and continued to eat and drink throughout all separations.

Discussion

The data derived from the second experiment support and extend the findings obtained in the first experiment. The animals were several months older when separated from their mothers, but it has been reported that age does not seem to be an important factor in modulating the response to separation (Coe, Glass, Wiener, & Levine, 1983; Mineka & Suomi, 1978). Although there was some reduction in the vocalization response of these older animals, this reduction appears to be a function of having prior separation experience, rather than a consequence of age (Gunnar et al., 1980).

In the adaptation phase of this experiment, in which mother and infant were housed together away from the social group, the infants did not appear to respond to this disruption, whereas the mothers showed an initial elevation of plasma cortisol immediately following removal from the social group. However, the mothers rapidly returned to basal levels, in contrast to our previously published experiments in which adult females without infants did not return to basal levels for several days following separation from their social groups (Gunnar et al., 1981).

General Discussion

Gunnar et al. (1980) reported that infants showed a small cortisol response and no behavioral signs of disturbance when removed from the social group in the presence of the mother; that finding is now replicated on two occasions in this study. This phenomenon of social buffering has been reported in squirrel monkey infants (Mendoza et al., 1978), and there are data available which indicate that social buffering also occurs in adult squirrel monkeys (Vogt, Coe, & Levine, 1981). By "social buffering," we mean the capacity of the presence of another organism or group of organisms within the environment to attenuate, and even eliminate, the plasma cortisol response that would normally appear when animals are exposed to novel or stressful stimuli. In this study, it is of interest that although the mother showed some aspects of this social buffering (as evidenced by the rapidity with which she returned to normal resting levels), it was the infant that seemed to derive the primary benefit. These data support the hypothesis that one of the primary functional consequences of mother–infant interactions is the capacity of the mother to provide the infant with the mechanisms by which the infant can modulate arousal.

Infants with visual access to the mother showed only a transient elevation of plasma cortisol, reduced in both magnitude and duration, compared with infants that were isolated in the presence of a peer. In this study, the response of the pituitary-adrenal system appeared to be in contrast to the infant's behavioral response. Vocalizations, whoo calls, and screams were emitted almost exclusively when the mother was present during separation. Mothers failed to show a cortisol response when separated in the presence of the infant but showed a significant elevation of plasma cortisol when completely isolated from the infant and in the presence of cues from their social group.

Vocalizations were influenced in this study by two factors: (a) the conditions of separation and (b) the duration of separation. Infants separated in the presence of their mothers vocalized significantly more than infants separated in the presence of peers. Furthermore, the rate of calling decreased over the 3-day mother-present separation condition reported here. Similar decrements in calling rates have been re-

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ported (Gunnar et al., 1981; Levine et al., 1984), and it seems likely that the time course reflects the infants' reevaluation of the effectiveness of the vocalization in restoring contact with the mother. In one sense, the decrement in vocalization is reminiscent of an extinction function. Thus, as the infant fails to obtain contact with the mother as a consequence of the signaling properties of vocalization, the emission of the response declines in relation to nonreinforcement. In the present experiment, those infants that had separation experience vocalized at a lower rate. This confirms the report by Gunnar et al. (1980) which indicated that prior separation experiences result in a reduced rate of vocalization. This evidence supports the notion that vocalization is a motor response which undergoes extinction as a consequence of prior experience with nonreinforcement. Thus, prior experience with nonreinforcement would lead to a rapid extinction. Similarly, we have found that in squirrel monkeys exposed to repeated separations, the rate of vocalization during the last separation is significantly reduced compared with that observed during first and second separations (Coe et al., 1983). These data appear to be consistent with the concept that vocalization is utilized by the infant as a means to achieve proximity and contact with the mother, which reduces arousal. The initial vocalizations following separation in the presence of the mother are an attempt to achieve proximity, and when unsuccessful, they tend to undergo diminution similar to that observed under any other condition of extinction for any learned function.

The infants' cortisol levels, though elevated shortly after separation from the mother, have been reported to return to basal levels (Gunnar et al., 1981), or at least to decline significantly (Gunnar et al., 1980; Levine et al., 1984), 24 hr later. Blood samples in those studies were taken in the late afternoon, and thus the findings are in agreement with the results of the present study in which afternoon cortisol levels in infants were at basal levels by the second day of separation. The current finding that cortisol remained elevated in the mornings of the second and third separation days in the mothers' absence is the first evidence of persistent elevation of cortisol lasting more than 24 hr in separated rhesus infants. In fact, the adrenocortical response was recurrent rather than persistent, as high levels were found in the mornings even when afternoon levels were basal. There are two possible explanations for this recurrent morning cortisol elevation. It may be that spending the night without the mother is more stressful than being without her during the day. Studies of infant development have found that even as infant monkeys mature and increase the time spent away from the mother, they still sleep with her at night. Reite and Short (1978) reported disruption of normal sleep patterns in separated rhesus infants. Alternatively, the pituitary-adrenal system may be more responsive in the morning than in the afternoon. Investigations on rats have indicated that the endocrinological response to stress may vary, depending upon the time of day that the stress is encountered (Stern, Goldman, & Levine, 1973).

Separation from the social group, and separation of mother and infant, affected mothers and infants in different ways. Mothers had greater cortisol elevations in response to capture than to separation. The only separation condition that was accompanied by a significant increase in cortisol was that during which the mothers were caged next to their social group. Thus, disruption of social bonds with group members appears to have been more arousing to these females than separation from their infants. Even in this condition, however, mothers' cortisol had returned to basal levels in the morning and afternoon of the second separation day. Infants had greater cortisol elevations when separated and isolated from their mothers than when removed from the social group and caged with the mother. The highest cortisol levels measured in rhesus monkeys in our laboratory were in animals removed both from their social group and from their mothers at the same time (Gunnar et al., 1980; Levine et al., 1984; Smotherman et al., 1979), perhaps indicative of a cumulative effect of these two stressful manipulations.

What emerges from these studies is a hierarchy of physiological responses to sep-

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aration conditions imposed on rhesus monkey infants. Separation from the social group into a novel environment while in contact with the mother appears to be minimally arousing; the infants show little evidence of behavioral agitation and no changes in plasma cortisol levels. Separation from the mother, but with visual or tactile access to her, results in an increase in vocalization, which declines as a function of length of separation, but the infants show little or no evidence of increased arousal as measured by changes in plasma cortisol level. However, higher cortisol levels are found in infants isolated from the mother in an unfamiliar environment. The most physiologically arousing situation is separation from the mother and from the social group simultaneously.

It is noteworthy that these differential responses of the pituitary-adrenal system are not reflected in the infants' overt behavior; in fact, the stimuli of the separation environment may affect the physiological and behavioral responses in a paradoxical manner. We have proposed that although seemingly paradoxical, these events are, in fact, highly related. Under conditions of stress, proximity and contact with the mother results in modulation and reduction of the infant arousal level, and the primary behavioral response that achieves proximity and contact in primates is vocalization. Thus, vocalization can function as a coping response on the part of the infant. When cues from the mother are available, the infant will vocalize in an attempt to achieve proximity. Although under conditions of the present experiments proximity and contact do not occur, it appears that the vocalization response still serves an arousal-reducing function, thus leading to either reduced levels of plasma cortisol or the elimination of the plasma cortisol response which frequently occurs under other conditions of separation.

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Psychological Documents to Be Discontinued

At its February 2–3, 1985, meeting, the Council of Representatives voted to cease publication of *Psychological Documents* (formerly the Journal Supplement Abstract Service) as of December 31, 1985, with the publication of the December 1985 issue of the catalog. Continued low submissions, decreasing usage, and rising costs for fulfillment of paper and microfiche copies of documents were reasons given for discontinuing publication of the alternative format publication, which was begun in 1971 as an "experimental" publication.

Authors who wish to submit documents for publication consideration must do so by July 1. Authors who are currently revising documents at the request of the editor should complete all revisions and submit them for final review as soon as possible, but no later than July 1. Orders for paper and microfiche copies of documents presently in the system and of those documents entered during 1985 will continue through December 31, 1986.